PROJECT DOCUMENTATION

AIR QUALITY ANALYSIS AND PREDICTION

INTRO - RESEARCH QUESTION

The goal of this project is to investigate daily particulate matter pollution (PM2.5 and PM10) across several monitoring stations, for this we combined station level daily air quality measurements with meteorological data. Our central research question: how do PM2.5 and PM10 concentrations vary across stations and over time, and which meteorological and pollutant variables are strongly associated with these concentrations? We also extend this into a predictive stage: can we forecast pollution levels, classify high pollution days, and identify natural pollution regimes.

1. DATA COLLECTION

Daily air-quality and meteorological data were collected for multiple monitoring stations, combining pollutant concentration measurements (PM2.5, PM10, and other gases such as NO, NO2, O3, SO2, CO) sourced from **OpenAQ**, with a wide set of daily-aggregated weather variables (temperature, humidity, wind, radiation, soil temperature/moisture, precipitation, and pressure) sourced from **Open-Meteo**.

2. DESCRIPTION OF DATA AND VARIABLES

Our working dataset is organized at the station day level: each row represents one station and one calendar date. Main variable groups:

- Identifiers: station_id, station_name, date.
- Target/pollutant variables: pm25, pm10 (daily average concentrations in μg/m³), plus other pollutants used as candidate predictors (no, no2, o3, so2, co).
- Coverage variables: an observed_hours column per pollutant, indicating how many of the 24 hourly readings were valid for that day's average.
- Meteorological variables: temperature (mean/min/max, 2m and apparent), relative humidity, dew point, surface and multi level pressure, precipitation and rain sums, snow depth, cloud cover, radiation variables, wind speed/gusts/direction, evapotranspiration, vapour pressure deficit, and soil temperature/moisture at several depths.
- Calendar variables: month, year, day of week, weekend indicator.

Only PM2.5 and PM10 are treated as targets. Other pollutants and meteorological variables are candidate explanatory features.

3. PREPROCESSING

We aggregated hourly measurements into daily averages, only treating a day as valid for a pollutant if enough hourly readings were available; rows with insufficient coverage were treated as missing rather than force filled. Before any further processing, we checked for duplicate (station_id, date) pairs to avoid double counting in later aggregations and models, and none remained unresolved. We then ran summary statistics (including tail percentiles), checked for negative values which are physically impossible, and discussed what to do with values above 500 μg/m³.

The most substantial check concerned physical consistency between PM10 and PM2.5. Since PM10 includes the PM2.5 size fraction, PM10 should always be at least as large as PM2.5 on the same station day. Out of 3,445 station days with both pollutants valid, 171 rows (about 5%) had PM10 below PM2.5. Most were small differences consistent with normal sensor noise, since PM2.5 and PM10 come from separate instruments. But 34 rows showed differences larger than 10 μg/m³, with the most extreme case reaching 94.5 μg/m³ (PM2.5 = 113.8, PM10 = 19.3). These were concentrated at specific stations, mainly Karpos and Centar, suggesting a station-specific issue rather than a general processing bug. We checked hourly coverage for these 34 rows: mean coverage was about 23.2 of 24 hours, essentially the same as other valid PM10 days, which ruled out low coverage as the explanation. Since we could not determine which pollutant reading was actually wrong, we did not delete or correct any values; instead we added a boolean quality flag (large_pm10_lt_pm25_flag) marking these rows, keeping the raw data intact and enabling an optional sensitivity check later. We also noted one isolated case of PM10 = 0.0 with valid PM2.5, likely a sensor fault, flagged for individual review.

Finally, we saved the cleaned daily table, including the quality flag, as a single Parquet file to serve as the single source of truth for all later notebooks.

4. ENCOUNTERED PROBLEMS

For PM10 falling below PM2.5 on some rows, we quantified the scale of the problem (about 5% of valid rows), separated small noise from large violations by a difference threshold, ruled out low hourly coverage as a cause, and documented the issue with a non-destructive flag instead of altering any values. In deciding how much effort to spend fixing this, we judged the modeling impact as small, since models treat PM2.5 and PM10 as independent columns unless a derived feature depends on their difference.

We feared that a derived difference column (pm_violation_diff = pm25 - pm10) would have created circular correlation with the targets if included in EDA, so we excluded it from the candidate features. The full correlation heatmap across all numeric columns was too dense to read, since many weather columns correlate strongly with each other, so we added a more focused analysis instead: predictors ranked by absolute correlation with PM2.5/PM10, a compact top-20 heatmap, and LOWESS scatterplots for the top six features. Some variables in an early monthly comparison plot were on very different scales, since CO's raw values were at a

larger scale than PM2.5, PM10, NO2, O3, and SO2, making the combined chart misleading. We noted this as a chart-design fix, to use separate axes or subplots, or normalize before plotting together.

5. METHODS USED

For the EDA stage, we used summary statistics with tail percentiles, since pollutant concentrations are right-skewed and the mean alone is misleading; histograms with KDE to visualize distribution shape and check for anomalies like spikes at zero; boxplots by station to compare medians and spread without letting outliers dominate; time-series line plots, both monthly means and daily values with a 30-day rolling average, to separate seasonal trend from day-to-day noise; scatterplots of PM2.5 vs PM10 and predictors vs targets to see relationship shape directly, including nonlinear patterns; and Pearson correlation for a fast linear-association screen, understanding it does not imply causation.

For feature engineering in notebook 06, we reindexed each station's series onto a complete calendar before computing time-based features, so lag and rolling values are computed over real calendar gaps rather than silently skipping missing days. We added three feature families: autoregressive lags at 1, 2, and 3 days plus rolling mean/std at 3 and 7 days, computed with shift(1) so no feature leaks the current day's value; cyclical month and day-of-week encodings using sine/cosine so December and January are treated as adjacent; and a binary high-pollution label, at 25 μg/m³ for PM2.5 and 50 μg/m³ for PM10, for the classification stage.

For regression forecasting in notebook 07, we compared Linear Regression, Random Forest, and Gradient Boosting against a persistence baseline, where today is predicted as yesterday's value, which is essential since pollution is highly autocorrelated day to day. We used a chronological train/test split, with the earliest 80% of dates for training and the most recent 20% for testing, rather than a random split, since random splits would leak future information into training.

For high-pollution classification in notebook 08, we reframed the problem as binary, PM2.5 above 25 μg/m³ or not, compared Logistic Regression, Random Forest, and Gradient Boosting against a majority-class baseline, and used precision, recall, F1, and ROC-AUC rather than relying on raw accuracy.

For clustering in notebook 09, we applied RobustScaler and PCA to the daily feature matrix, then compared KMeans, a Gaussian Mixture Model, and DBSCAN (eps/min_samples tuned by grid search). We set the cluster count at k = 4, the smallest k whose highest-pollution cluster was winter-dominant, rather than taking the silhouette optimum at k = 2, which merely produced a season-neutral catch-all cluster."

6. WHAT WE LEARNED FROM THE EDA

Both PM2.5 and PM10 are strongly right-skewed, with a long tail of high-pollution days. PM10 has a wider range, consistent with including both fine and coarse particles. Medians and percentiles are therefore more representative than the mean alone.

Median pollution differs meaningfully by station. Karpos and Centar had the highest typical PM2.5, Rektorat and Centar had the highest PM10, and Miladinovci was generally lowest for both. This makes station identity a strong candidate feature.

Pollution is highest in December and January and lowest from roughly April to September, with both pollutants moving together across the year. Month or season should be included as a feature. Winter conditions such as heating, temperature inversions, and reduced dispersion are a likely driver, though EDA only shows association rather than proof.

The multi-year daily plot confirmed the same seasonal cycle repeating every year, with sharp short-term spikes from individual pollution episodes. A flat-looking recent segment was noted as needing a coverage check rather than being read as genuinely stable air quality.

The PM2.5-versus-PM10 relationship is clearly positive, with most points sitting on or above the "PM10 = PM2.5" line as expected. The flagged inconsistent rows sit clearly below the line, confirming that the flag captures genuine outliers rather than ordinary variation.

In the correlation screening, CO (r about 0.51 with PM2.5, 0.62 with PM10) and NO (about 0.47 with both) were the strongest associated variables. This plausibly reflects shared combustion and traffic sources. Temperature-related variables showed a consistent negative association, roughly -0.40 to -0.48, which lines up with the seasonal pattern already seen. Radiation and evapotranspiration showed similar negative associations, likely acting as season or sunlight proxies rather than direct drivers. O3 was moderately negative, around -0.35, and SO2 was weakly positive, between 0.18 and 0.34. The derived pm_violation_diff column was excluded from this screening as circular.

Overall, three variable groups stand out as promising features for modeling: station identity, calendar or season, and a small set of correlated pollutants and temperature-family weather variables such as CO, NO, temperature, and soil temperature. Wind, humidity, and radiation showed weaker linear correlation but may still help nonlinear models pick up patterns that correlation alone cannot capture.

7. REGRESSION FORECASTING RESULTS

Linear Regression performed best, with RMSE 13.62, MAE 8.93, and R^2 0.734. This narrowly beat the persistence baseline of RMSE 14.58, MAE 8.67, and R^2 0.695. Random Forest (RMSE 15.16) and Gradient Boosting (RMSE 18.05) both underperformed the naive baseline.

This is a genuinely informative negative result. The more complex models overfit or failed to generalize, while the simplest linear model captured the dominant linear relationship more reliably. For this model, soil moisture and snow depth were shown as the most important features. This likely reflects seasonal confounding, since snow and moisture act as winter proxies rather than direct physical drivers. Lag and rolling pollution features contributed comparatively little.

8. HIGH-POLLUTION CLASSIFICATION RESULTS

The majority-class baseline reached 68.1% accuracy purely from class imbalance, since only about 33.6% of days are labeled "high." Random Forest was the best performer, with Accuracy 0.893, Precision 0.768, Recall 0.951, F1 0.850, and ROC-AUC 0.965. This was ahead of Logistic Regression (F1 0.838) and Gradient Boosting (F1 0.824), and far ahead of the majority baseline (F1 0.000).

The high recall of 0.951 is particularly valuable for a pollution-warning use case, since missing a real high-pollution day is costlier than a false alarm. Feature importance here showed lagged and rolling pollution values as the strongest predictors, with soil and temperature features secondary. This is a sensible result, since recent pollution levels strongly predict near-term levels.

9. CLUSTERING RESULTS AND IMPROVEMENTS

We clustered the 1,912-row modelling matrix using 21 features: autoregressive lags, rolling means/std, cyclical month/day-of-week encodings, core weather variables, a rain indicator, an is-weekend flag, and wind-category dummies. Continuous features were scaled with RobustScaler and reduced to 2 PCA components capturing 90% of the variance.

Silhouette score alone favored k=2, but that solution was misleading: its "high-pollution" cluster absorbed most rows and split almost evenly between summer and winter, earning a "summer" label. Since EDA's clearest signal was the December/January peak, we instead chose the smallest k whose top cluster was genuinely winter-dominant, landing on k=4.

We compared KMeans, a Gaussian Mixture Model, and DBSCAN, the latter added as a density-based method that doesn't require specifying k and can flag ambiguous points as noise. KMeans and DBSCAN found well-separated structure; GMM's clusters overlapped more and were harder to distinguish.

All three methods agreed on the core story: a compact, cold-season, high-PM2.5 regime cleanly separates from a broader, season-neutral band of typical days. KMeans and GMM each added a few smaller autumn-leaning intermediate clusters without disturbing this main split. DBSCAN's distinct contribution was setting aside a small set of extreme, mostly-winter days as noise rather than folding them into a general cluster.

Overall, the clustering now aligns with the seasonal pattern from EDA, supporting a real cold-season pollution regime rather than an artifact of scaling or cluster-count choice.

10. CONCLUSION (FOR NOW)

Skopje's PM2.5 problem is largely seasonal, daily averages peak in December and January, with CO, SO2, and NO showing the strongest positive associations, and temperature and ozone the strongest negative ones. With a third of all days exceeding the 25 μg/m³ threshold, this is a chronic problem rather than an occasional spike.

For forecasting, no model clearly dominates. Linear regression (RMSE 13.6, R^2 0.734) edges out a simple lag-1 persistence baseline (RMSE 14.6, R^2 0.695), while Random Forest (RMSE 15.2) and Gradient Boosting (18.1) both underperform it. The daily signal is largely linear, driven by inherited pollution and temperature, with snow depth and soil moisture standing out mainly as winter proxies.

For classifying high-pollution days, Random Forest is the clear winner: 89.3% accuracy, 95.1% recall, F1 0.850, and ROC-AUC 0.965, well ahead of logistic regression (F1 0.838), gradient boosting (0.824), and the majority-class baseline (68.1% accuracy, F1 0).

Clustering at k=4 confirms the seasonal story. KMeans isolates a distinct winter-smog regime, 358 days averaging 50.9 μg/m³ and 81% winter, against a much larger moderate background. GMM finds a softer version of the same split, while DBSCAN flags about 5% of days as noise, capturing the most extreme, cold-season outliers. The overall picture isn't a smooth pollution continuum: most days are moderate, and a reproducible winter regime is the pattern most worth targeting for early-warning classification and mitigation.